

An Introduction to Entropy and its Applications in Combinatorics

Why and how to turn a counting problem into an uncertainty problem

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August 5, 2026

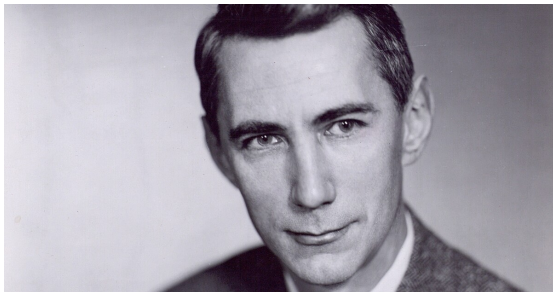


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*"My greatest concern was what to call it. I thought of calling it 'information,' but the word was overly used, so I decided to call it 'uncertainty.' When I discussed it with John von Neumann, he had a better idea. Von Neumann told me, 'You should call it **entropy**, for two reasons. In the first place your uncertainty function has been used in statistical mechanics under that name, so it already has a name. In the second place, and more important, no one really knows what entropy really is, so in a debate you will always have the advantage.'"* - **Claude Shannon, Father of Information Theory**



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- ▶ Entropy gives us the right language/framework for thinking about certain problems which can help us with discovering/verifying the solution.
- ▶ It comes with a prepackaged toolkit which saves us the effort of re-deriving equivalent results. Moreover, not all of these tools have a direct counting equivalent.

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- ▶ **2000s to today:** Rapid expansion in the use of entropy in combinatorics.

1

An Introduction to Entropy

How should entropy be defined?

Summarised from *Three tutorial lectures on entropy and counting* (David Galvin 2014):

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1. $S(1) = 0$ (no surprise in seeing a certain event)
2. If $p < q$, then $S(p) > S(q)$ (rarer events are more surprising)
3. S varies continuously with p
4. $S(pq) = S(p) + S(q)$ (the surprise of observing two independent events)
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Claim: The unique function satisfying the above conditions is $S(p) = -\log_2 p = \log_2 \frac{1}{p}$.

Formally defining entropy

The entropy of a discrete random variable

Let X be a discrete random variable which takes on values $x_1, \dots, x_m$, with probabilities $p_1, \dots, p_m$ respectively. The **entropy** of X , denoted by $H(X)$, is given by

$$H(X) := \sum_{i=1}^m -p_i \log_2 p_i = \sum_{i=1}^m p_i \log_2 \frac{1}{p_i}.$$

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- ▶ Observe that the entropy of a discrete random variable depends only on the probability of the outcomes, and not on the actual value of the outcomes.
- ▶ If we let p_X denote the probability mass function of X , we have

$$H(X) = \mathbb{E}[-\log_2 p_X(X)] = \mathbb{E}[S(p_X(X))].$$

That is, entropy is just the expected/average surprise of X .

Example 1: the entropy of a fair coin

Let X be the outcome of a fair coin flip. Then the two outcomes, heads and tails, each occur with probability $1/2$. Thus, the entropy of X is given by

$$H(X) = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1 \text{ bit.}$$

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That is, from an information theory point of view, observing the outcome of a (fair) coin flip gives you 1 bit of information on average.

Example 2: the entropy of a biased coin

Let Y be the outcome of a coin flip where heads occurs 90% of the time and tails occurs only 10% of the time. Then the entropy of Y is given by

$$H(Y) = -\frac{1}{10} \log_2 \frac{1}{10} - \frac{9}{10} \log_2 \frac{9}{10} \approx 0.469 \text{ bits.}$$

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We emphasise that entropy measures the **average** uncertainty of a random variable.

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Let Z be the outcome of a coin flip with two heads. Then the entropy of Z is given by

$$H(Z) = -0 \log_2 0 - \log_2 1 = 0 \text{ bits},$$

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The lesson here is that entropy is maximised when each of the possible outcomes is equally likely. In other words,

$$H(X) \leq \log_2 |\text{supp}(X)|,$$

with equality occurring when X is uniform in its support. This is one of the key properties of Shannon's definition of entropy and is what makes it useful in the context of counting.

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- ▶ In this way, any bound we obtain on $H(X)$ ultimately implies a corresponding bound on $|\mathcal{F}|$. (What happens if X is not uniformly chosen from $\mathcal{F}$?)
- ▶ **Question:** how do we actually bound $H(X)$? Why would it be easier to bound $H(X)$ instead of directly trying to bound $|\mathcal{F}|$?

2

The Entropy Toolkit

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- ▶ So far, we have only considered the entropy of a single random variable X .
- ▶ Entropy is especially powerful when one considers a random vector $(X_1, \dots, X_n)$, which itself is a random variable whose probability distribution is just the joint distribution of its coordinates.
- ▶ That is, if $\mathbf{X} = (X_1, \dots, X_n)$, then

$$H(\mathbf{X}) = H(X_1, \dots, X_n) = \sum_{x_1, \dots, x_n} -p_{\mathbf{X}}(x_1, \dots, x_n) \log_2 p_{\mathbf{X}}(x_1, \dots, x_n),$$

where

$$p_{\mathbf{X}}(x_1, \dots, x_n) = \Pr(X_1 = x_1, \dots, X_n = x_n).$$

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- ▶ We omit the proof for now but it follows easily from the results we introduce next.

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Let $\mathcal{F}$ be a family of subsets of $[n]$ (possibly with repeats) with each $i \in [n]$ included in at least r members of $\mathcal{F}$. For a random vector $(X_1, \dots, X_n)$, we have

$$r \cdot H(X_1, \dots, X_n) \leq \sum_{F \in \mathcal{F}} H(X_F),$$

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- ▶ Basically, if we have a family of overlapping local views that sees every coordinate at least r times, then together they contain at least r copies worth of the information in the whole object.
- ▶ We will prove the three variable case after introducing some other properties of entropy.

Simple example of Shearer's lemma

Let X be the result of a fair coin flip and suppose $X_1 = X_2 = X_3 = X$ so that the random variables are completely dependent. Consider the random vector (X_1, X_2, X_3) whose entropy is 1 bit.

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$$H(X_1, X_2, X_3) \leq H(X_1) + H(X_2) + H(X_3) = 3 \text{ bits.}$$

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Applying Shearer's lemma with $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$, we have

$$2H(X_1, X_2, X_3) \leq H(X_1, X_2) + H(X_1, X_3) + H(X_2, X_3)$$

which implies

$$H(X_1, X_2, X_3) \leq 3/2 \text{ bits.}$$

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This simple example demonstrates that, in principle, Shearer's lemma can give you a better bound than subadditivity.

Conditional entropy

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Conditional entropy

For an event E , we define the entropy of X given E as

$$H(X|E) = \sum_x -p(x|E) \log_2 p(x|E).$$

For a pair of random variables X and Y , we define the entropy of X given Y as

$$H(X|Y) = \mathbb{E}_Y[H(X|\{Y = y\})] = \sum_y p(y) \sum_x -p(x|y) \log_2 p(x|y).$$

Here, $\mathbb{E}_Y$ means we are taking the expectation with respect to the random variable Y .

Chain rule

The chain rule shows that the entropy of a random vector can be decomposed into the sum of successive conditional entropies of its coordinates.

Chain rule

For random variables X and Y , we have

$$H(X, Y) = H(X) + H(Y|X).$$

More generally, for random variables $X_1, \dots, X_n$, we have

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1}).$$

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- ▶ Suppose we roll an 8-sided die. Let X be the number rolled and let Y be the parity of the number rolled. It is not hard to see that $H(X) = 3$ bits and $H(Y) = 1$ bit.

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- ▶ If we first observe Y , then this halves the space of outcomes for X so that $H(X|Y) = 2$ bits. Thus, by the chain rule we have

$$H(X, Y) = H(Y, X) = H(Y) + H(X|Y) = 1 + 2 = 3 \text{ bits.}$$

- ▶ Observe that in this scenario, subadditivity only gives us

$$H(X, Y) \leq H(X) + H(Y) = 4 \text{ bits.}$$

Proof of the chain rule

We prove the two variable case. The general case can then be handled by induction.

$$H(X, Y) - H(X) = \sum_{x,y} -p(x, y) \log_2 p(x, y) - \sum_x -p(x) \log_2 p(x)$$

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Basically, knowing the outcome of Y (or Z) should not, **on average**, increase our surprise when learning the outcome of X . In other words, having more information should never hurt us overall.

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For random variables X, Y, Z , we have

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The proof for the general case is similar but requires more bookkeeping.

3

Applications

Application 1: The Loomis–Whitney inequality

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$$|B| \leq \prod_{j=1}^n |B_j|^{1/(n-1)}.$$

It turns out that this bound is tight. For instance, one has equality when B is an axis-aligned (hyper)cube.

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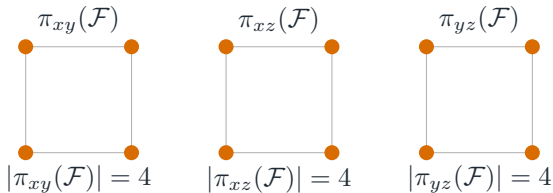
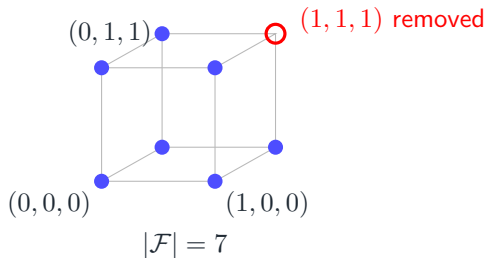
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It turns out that the discrete Loomis-Whitney inequality admits a direct proof using Shearer's lemma.

Discrete Loomis–Whitney inequality example

$$\mathcal{F} = \{0, 1\}^3 \setminus \{(1, 1, 1)\}$$

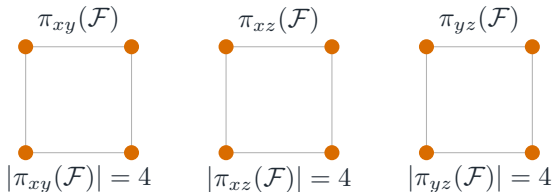
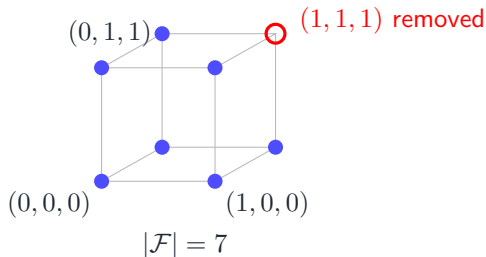


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The true size of $\mathcal{F}$ is 7 but the Loomis-Whitney inequality gives $|\mathcal{F}| \leq 8$ since the missing corner is invisible in every two-dimensional projection.

Proof of the discrete Loomis-Whitney inequality

Choose $\mathbf{X} = (X_1, \dots, X_n)$ uniformly randomly from $\mathcal{F}$. We seek to apply Shearer's lemma with the coordinate sets $[n] \setminus \{1\}, \dots, [n] \setminus \{i\}, \dots, [n] \setminus \{n\}$.

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Main idea: Since every coordinate appears in all but one shadow, they collectively contain at least $n - 1$ copies worth of the information in the original point.

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- ▶ Unlike the determinant, there is no simple geometric interpretation of the permanent.

Bregman's theorem and perfect matchings

Bregman's theorem (Bregman-Minc inequality)

Let $A = (a_{ij}) \in \{0, 1\}^{n \times n}$ be an $n \times n$ binary matrix. If we let d_i denote the sum of the i -th row then

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- ▶ **Claim:** $\text{per}(A) = \text{number of perfect matchings in a bipartite graph } G$.

Bipartite graph example

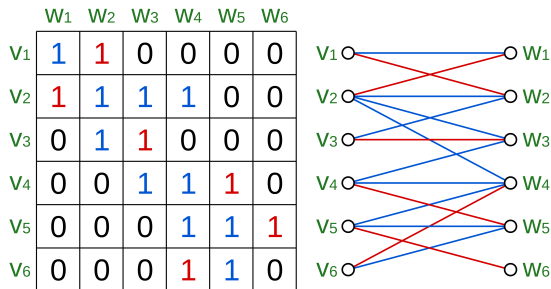


Figure 1: A bipartite adjacency matrix and the corresponding bipartite graph with a possible perfect matching marked in red. According to Bregman's theorem, there are at most 20 perfect matchings in this graph.¹ Note that equality is attained in Bregman's theorem when the bipartite adjacency matrix consists of diagonal blocks of 1's.

¹Image and caption sourced from the Wikipedia page on the Bregman–Minc inequality

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- ▶ Hence, the sum over all $\sigma \in S_n$ for which $\prod_{i=1}^n a_{i\sigma(i)} = 1$ counts exactly the number of perfect matchings.

Proof of Bregman's theorem (naive attempt)

Fix some $A = (a_{ij}) \in \{0, 1\}^{n \times n}$. Let $\mathcal{M} = \{\sigma \in S_n \mid \prod_{i=1}^n a_{i\sigma(i)} = 1\}$ denote the set of valid permutations. Let σ be a valid permutation sampled uniformly randomly from $\mathcal{M}$.

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$$\log_2 \text{per}(A) = \log_2 |\mathcal{M}| = H(\sigma) = H(\sigma_1) + H(\sigma_2|\sigma_1) + \dots + H(\sigma_n|\sigma_1, \dots, \sigma_{n-1}).$$

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From here, we can try dropping conditioning to obtain

$$H(\sigma_i|\sigma_1, \dots, \sigma_{i-1}) \leq H(\sigma_i) \leq \log_2 |\text{supp}(\sigma_i)| \leq \log_2 d_i.$$

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The issue is that by dropping conditioning, we are pretending each row has d_i choices when in reality, the number of choices is successively decreasing.

A working example

Consider the binary matrix

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

For row 4, the allowable columns are $\{1, 2, 4, 6\}$ and we have $d_4 = 4$. Thus, $H(\sigma_4) \leq \log_2 4 = 2$ bits.

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Suppose $\sigma_1 = 1, \sigma_2 = 2, \sigma_3 = 3$. If we reveal the rows in order, then conditional on $\sigma_1 = 1, \sigma_2 = 2, \sigma_3 = 3$, the support of σ_4 is $\{4, 6\}$ and thus, by the uniform bound, $H(\sigma_4 | \sigma_1, \sigma_2, \sigma_3) \leq \log_2 2 = 1$ bit.

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By dropping conditioning, we are overestimating the true entropy of σ_4 at the time of revealing.

Proof of Bregman's theorem (Radhakrishnan 1997)

The following is based on Yufei Zhao's notes in *Probabilistic Methods in Combinatorics*:

- ▶ **The key insight from Radhakrishnan was to reveal the chosen entries in a uniform random order.** We now additionally consider a uniformly random permutation $\tau \in S_n$ which represents the order in which we reveal the rows of the matrix. **This is independent of σ .** We say that i appears before j if $\tau_i < \tau_j$.

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- ▶ Let $N_i = N_i(\sigma, \tau)$ be the number of ones in row i that do not lie in the same column as some entry (j, σ_j) that comes before i . Here, N_i can be interpreted as the number of “greedily available” choices for σ_i before it is revealed (the available columns j for which $a_{ij} = 1$ but may not necessarily extend into a perfect matching).

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- ▶ For each fixed τ , the chain rule gives us

$$H(\sigma) = \sum_{i=1}^n H(\sigma_i | \sigma_j : j \text{ comes before } i)$$

and the uniform bound gives

$$H(\sigma_i | \sigma_j : j \text{ comes before } i) \leq \mathbb{E}_\sigma [\log_2 N_i],$$

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 - ▶ Row i has d_i allowable choices so exactly d_i rows have their chosen column among these allowable columns (including row i itself). That is, $|S_i| = d_i$.
 - ▶ Since τ reveals the rows uniformly randomly, the key observation is that the relative order of the d_i elements of S_i is also uniformly random. Hence, each possible relative position for row i (from first to d_i -th) occurs with probability $1/d_i$. Moreover, the relative position of when row i is revealed determines exactly the size of N_i .

Returning to our working example

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ \color{red}{1} & \color{red}{1} & \color{red}{0} & \color{red}{1} & \color{red}{0} & \color{red}{1} \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}, \quad \sigma = (1, 2, 3, 4, 5, 6).$$

We again focus on row 4 whose initial allowable columns are $\{1, 2, 4, 6\}$.

Rows already revealed	Used columns	Available columns for row 4	N_4
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- ▶ Thus, we have

$$\mathbb{E}_{\tau}[\log_2 N_4 | \sigma] = \frac{\log_2 1 + \log_2 2 + \log_2 3 + \log_2 4}{4} = \frac{\log_2 4!}{4} \approx 1.146 \text{ bits.}$$

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For any fixed σ , the same argument applies to the corresponding d_4 -element set $S_4(\sigma)$.

Proof of Bregman's theorem (Radhakrishnan 1997)

- By the previous claim, we have that for each fixed σ ,

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- Thus, taking the expectation over σ and summing over i , we have

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- Exponentiating, we obtain

$$\text{per}(A) \leq \prod_{i=1}^n (d_i!)^{1/d_i}.$$

4

Closing

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5. Combine inequalities/take exponents to get a bound on $|\mathcal{F}|$.

$$\log_2 |\mathcal{F}| = H(X) \leq B \implies |\mathcal{F}| \leq 2^B.$$

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2. Here, each X is naturally of the form $(X_1, \dots, X_n)$ where $X_i \in A_i$.
3. The local views we consider are the $(n-1)$ -dimensional projections $X_{[n] \setminus \{i\}}$ and we apply Shearer's lemma to obtain

$$(n-1)H(X) \leq \sum_{i=1}^n H(X_{[n] \setminus \{i\}}).$$

4. For each local view, we have the bound $H(X_{[n] \setminus \{i\}}) \leq \log_2 |\pi_i(\mathcal{F})|$.
5. Combining the above inequalities and taking exponents, we have

$$|\mathcal{F}|^{n-1} \leq \prod_{i=1}^n |\pi_i(\mathcal{F})|.$$

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5. Combining the above inequalities, we have

$$\log_2 \text{per}(A) = H(\sigma) \leq \sum_{i=1}^n \mathbb{E}_{\sigma, \tau}[\log_2 N_i] = \sum_{i=1}^n \mathbb{E}_\sigma[\mathbb{E}_\tau[\log_2 N_i | \sigma]] = \sum_{i=1}^n \frac{\log_2(d_i!)}{d_i}.$$

Exponentiating yields $\text{per}(A) \leq \prod_{i=1}^n (d_i!)^{1/d_i}$.

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- ▶ Yet, even with the limited concepts/tools we have, we could understand two landmark entropy-based proofs in combinatorics.
- ▶ Of course, there are many more beautiful applications of entropy in combinatorics which we did not get to see today.

Resources and Further Reading

The following resources were extremely helpful in the preparation of these slides and I recommend checking them out:

- ▶ *Three tutorial lectures on entropy and counting* (Galvin 2014)
- ▶ *Probabilistic Methods in Combinatorics* (Yufei Zhao)
- ▶ *Entropy Methods in Combinatorics* (Samotij 2026)

Thank you

Questions?